

# Heme Synthesis & Porphyrrias - Factoid Sheet

Week 7 Hematology • rapid post-lecture review • companion to Browning slide notes

**Official LO focus:** structure/function + pathway; liver vs marrow regulation; AIP/PCT/EPP/lead; rationale for glucose/hemin. **ScholarRx: Heme Metabolism**

## Pathway - memorize this map

| # | Enzyme                     | Product                           | Where / hook                                 |
|---|----------------------------|-----------------------------------|----------------------------------------------|
| 1 | <b>ALA synthase</b>        | Glycine + succinyl-CoA -> ALA     | Mito; <b>rate-limiting</b> ; B6; ALAS1/ALAS2 |
| 2 | <b>ALA dehydratase</b>     | ALA -> PBG                        | Cytosol; Zn; <b>lead</b>                     |
| 3 | <b>PBG deaminase/HMBS</b>  | PBG -> hydroxymethylbilane        | Cytosol; <b>AIP</b>                          |
| 4 | UPG III synthase           | Hydroxymethylbilane -> UPG III    | Cytosol                                      |
| 5 | <b>UPG decarboxylase</b>   | UPG III -> coproporphyrinogen III | Cytosol; <b>PCT</b>                          |
| 6 | Coproporphyrinogen oxidase | -> protoporphyrinogen IX          | Mito                                         |
| 7 | Protoporphyrinogen oxidase | -> protoporphyrin IX              | Mito                                         |
| 8 | <b>Ferrochelatase</b>      | + Fe <sup>2+</sup> -> heme        | Mito; <b>EPP + lead</b>                      |

### Location mnemonic

**Mito starts -> cytosol middle -> mito finishes.**  
Faculty handout: step 1 + final 3 are mitochondrial.

### Intermediate rule

**Early ALA/PBG -> neurovisceral.**  
**Later porphyrins -> photosensitivity.**

## Regulation - ALAS1 vs ALAS2

### ALAS1 (hepatic)

- Heme negative feedback
- Fasting/glucagon -> PGC-1alpha -> ↑ ALAS1
- CYP450-inducing drugs/hormones -> ↑ ALAS1
- Insulin/carbohydrate suppress fasting drive

### ALAS2 (erythroid)

- EPO -> JAK/STAT5 -> ↑ transcription
- Iron/IRE controls translation
- Low iron -> IRP blocks ALAS2 translation
- Heme production coordinated with globin

**LO 4 mechanism:** Glucose reduces fasting-driven hepatic ALAS1. Hemin restores heme feedback inhibition of ALAS1. Both reduce ALA/PBG production upstream of an AIP block.

# Disease Factoids - If You See X, Think Y

The four conditions explicitly named in the official learning objectives

|                  | AIP                                       | PCT                                              | EPP                                                      | Lead                                                    |
|------------------|-------------------------------------------|--------------------------------------------------|----------------------------------------------------------|---------------------------------------------------------|
| <b>Defect</b>    | PBG deaminase / HMBS                      | UROD activity                                    | Ferrochelatase                                           | ALAD + ferrochelatase inhibited                         |
| <b>Build-up</b>  | ALA + PBG                                 | Uroporphyrins                                    | Protoporphyrin IX                                        | ALA + zinc/RBC protoporphyrin                           |
| <b>Signature</b> | <b>Neurovisceral; NO photosensitivity</b> | <b>Delayed blistering</b> photosensitivity       | <b>Immediate painful NON-blistering</b> photosensitivity | GI/neuro symptoms + microcytic hypochromic anemia       |
| <b>Test</b>      | Urine PBG                                 | Urine uroporphyrin                               | Blood/RBC protoporphyrin                                 | Blood lead definitive; Zn-protoporphyrin supportive     |
| <b>Tx clue</b>   | Remove trigger; glucose; hemin            | Triggers/light; phlebotomy; low-dose chloroquine | Light; beta-carotene; afamelanotide                      | Remove exposure; severity-directed toxicology treatment |

## Vignette triggers

**Severe abdominal pain + confusion/psych + neuropathy after fasting or phenobarbital**  
-> AIP. Think ↑ urine PBG; treat by suppressing ALAS1.

**Adult with fragile blistering dorsal hands + scarring/hypertrichosis**  
-> PCT. Think UROD + urine uroporphyrin + HCV/iron/ alcohol associations.

**Child burns/swells within minutes of sun, no blisters**  
-> EPP. Think ferrochelatase + RBC protoporphyrin + hepatobiliary risk.

**Abdominal/neuro symptoms + anemia after lead exposure**  
-> ALAD + ferrochelatase inhibition; ↑ ALA and Zn-protoporphyrin.

## Professor-specific details worth knowing

- **PCT sunscreen nuance:** porphyrins absorb visible light, so ordinary UV-only sunscreen may be inadequate; faculty handout favors opaque physical blockers/protective clothing.
- **EPP urine trap:** protoporphyrin IX is water-insoluble -> bile/feces, not primarily urine.
- **EPP liver:** gallstones/cholestasis; rust-brown hepatic deposits; Maltese-cross birefringence in severe hepatic disease.
- **Lead vs EPP:** both involve ferrochelatase, but zinc protoporphyrin is less phototoxic, so lead does not mimic classic EPP photosensitivity.
- **Porphyrinogen vs porphyrin:** porphyrinogens are colorless; oxidation generates colored porphyrins.

## Board enrichment / common add-ons

**Lead:** basophilic stippling, ring sideroblasts, wrist/foot drop may appear in board-style stems. **ALAS:** requires B6, linking B6 deficiency/isoniazid to impaired heme synthesis and sideroblastic anemia.

**Last 5 minutes:** AIP=PBGD=neurovisceral=urine PBG=glucose/hemin • PCT=UROD=adult blistering=urine uroporphyrin • EPP=ferrochelatase=child immediate burning/no blisters=RBC protoporphyrin • Lead=ALAD+ferrochelatase=ALA+Zn-protoporphyrin+microcytic anemia.

Primary lecture basis: Browning 2026 slides + faculty handout. Official Week 7 objectives set the scope. Separate slide-companion notes contain full explanations.